

Abstract

Background: Premenstrual dysphoric disorder (PMDD) is a DSM-5 depressive disorder affecting an estimated 1.6% of women and girls (~31 million) worldwide.¹ This scoping review mapped psychosocial outcomes and coping mechanisms among women with PMDD and the geographic distribution of the evidence.

Methods: Following Arksey and O'Malley's framework and PRISMA-ScR, records from CINAHL, APA PsycInfo, APA PsycArticles, MEDLINE, Women's Studies International, PubMed and the Cochrane Library (2010–2025) were screened. Eligible peer-reviewed empirical or qualitative studies had PMDD as the population and a psychosocial outcome or coping mechanism as a primary aim.

Results: Of 1,801 records, 342 were duplicates; 1,459 were screened and 85 studies were included. Most studied were depression (n=28), mood and emotion regulation (n=18), psychiatric comorbidity (n=18), psychological distress (n=16) and suicidal ideation or self-harm (n=13); only 7 studies examined coping. Two-thirds (67%) came from high-income countries; 3 (3.5%) came from sub-Saharan Africa (Ethiopia, n=2; Nigeria, n=1).

Conclusions: Evidence on the psychosocial burden of PMDD is growing but is concentrated in high-income settings, relies on provisional diagnoses and rarely addresses coping.

Keywords: premenstrual dysphoric disorder; psychosocial outcomes; coping mechanisms; LMICs; scoping review

Introduction, Objective, and Research Question

Introduction: PMDD is a recurring, hormone-linked mood disorder in which irritability, mood swings, depressed mood and anxiety emerge in the late luteal phase and remit after menses begin; DSM-5 classifies it as a depressive disorder.² Using prospectively confirmed diagnoses, community prevalence is about 1.6%, roughly 31 million women and girls worldwide.¹ PMDD has been linked with suicidal thoughts, strained relationships and reduced quality of life, but this evidence is scattered across disciplines, and it is unclear how women cope or where the research has been conducted.

Objective: To map psychosocial outcomes and coping mechanisms reported among women with PMDD and identify geographic gaps in the published literature.

Research Question: What psychosocial outcomes and coping mechanisms have been reported in studies of women with PMDD published between 2010 and 2025, and how are these studies distributed across geographic settings?

Methods

Category	Description
Study Design	Scoping review guided by Arksey and O'Malley's framework ³ and reported using PRISMA-ScR. ⁴
Databases	CINAHL Ultimate, APA PsycInfo, APA PsycArticles, MEDLINE Ultimate, Women's Studies International, PubMed/MEDLINE and Cochrane Library.
Publication Years	2010–2025; English language.
Inclusion Criteria	Peer-reviewed empirical or qualitative studies with PMDD (provisional or confirmed) as the study population and ≥1 psychosocial outcome or coping mechanism as a primary aim.
Exclusion Criteria	Animal, laboratory/biological, intervention, review, instrument-validation and case-report studies; non-peer-reviewed publications; PMS-only samples.
Screening Process	1,801 records deduplicated (DOI and title matching) to 1,459; titles, abstracts and subject terms screened with every exclusion reason logged; 85 studies included.
Evidence Mapping	Studies charted by psychosocial domain (13 categories), coping mechanism, PMDD diagnostic method, country, World Bank region and income group (FY2027).

1,801

records identified
across 7 databases

85

studies included
in the scoping review

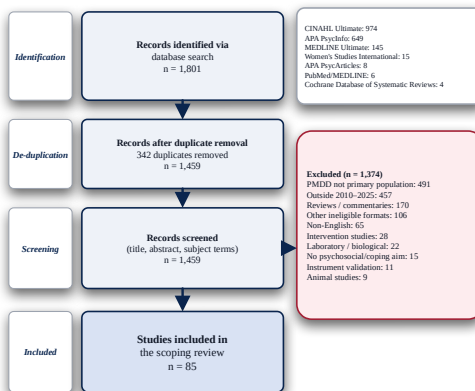
7

studies addressing coping
8.2% of 85

3

studies from sub-Saharan Africa
3.5% of 85

Fig 1. Study Selection — PRISMA-ScR Flow Diagram

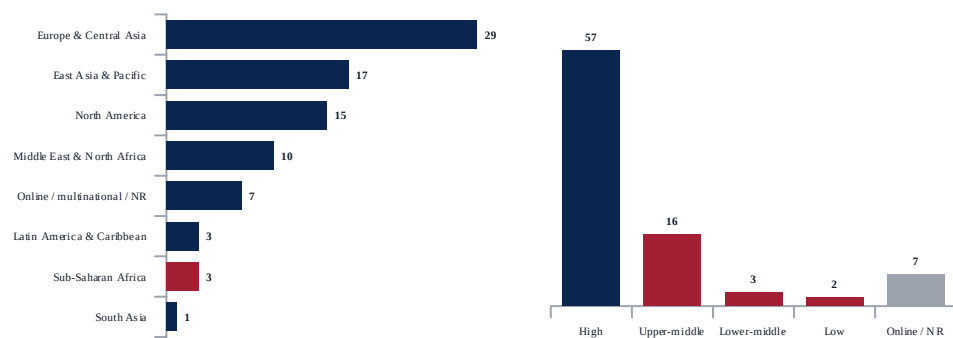


Eligibility judged on titles, abstracts and subject terms; full-text verification ongoing. All decisions logged in the supplementary workbook.

Inclusion Criteria (Arksey & O'Malley, 2005 — scoping review framework)

- ✓ Peer-reviewed empirical or qualitative studies, 2010–2025
- ✓ PMDD as the primary population
- ✓ At least one psychosocial outcome or coping mechanism as a primary aim

Fig 3. Share of the 85 Included Studies, by World Bank Region and Income Group



Sub-Saharan Africa: 3 studies (Ethiopia 2, Nigeria 1). Red columns = low- and middle-income countries (21 studies, 25%); only 5 (5.9%) from low or lower-middle-income countries. Income groups: World Bank FY2027.

Results, Discussion, Conclusion, and Limitations

Results

Of 1,801 records, 342 duplicates were removed; 1,459 titles and abstracts were screened and 85 studies met the inclusion criteria (Fig. 1). Nearly half (49%) were published since 2020; most were cross-sectional (56%), and only 24% confirmed PMDD with prospective daily ratings.

Depression was the most frequently examined outcome (n=28), followed by mood and emotion regulation and psychiatric comorbidity (n=18 each), psychological distress (n=16) and suicidal ideation or self-harm (n=13) (Fig. 2). PMDD was associated with suicidal ideation (OR 2.22) and attempts (OR 2.10) in a US national sample,⁵ and 72% of a global PMDD sample reported lifetime active suicidal ideation.⁶ Only 7 studies examined coping, mostly self-medication or maladaptive coping styles.

Two-thirds of studies (67%) came from high-income countries; 3 came from sub-Saharan Africa (Fig. 3).

Discussion

The literature consistently links PMDD with depression, suicidality, impaired functioning and reduced quality of life, but it relies heavily on provisional diagnoses, and coping and lived experience remain understudied. No study from sub-Saharan Africa examined suicidality, quality of life or coping, so the burden there is largely undocumented.

Conclusion
PMDD research documents a substantial psychosocial burden, while coping mechanisms and low-income settings remain underrepresented. Culturally responsive screening and Africa-centred research are needed.

Limitations

- **Screening stage:** eligibility judged on titles and abstracts; full-text verification ongoing.
- **Search scope:** English only; African Journals Online and grey literature not searched.
- **Diagnosis:** many studies used provisional diagnoses or combined PMS and PMDD.
- **Coding overlap:** studies can address several domains, so counts should not be summed.

Public Health Implications and Future Work

Public Health Implications

- **Screen for suicide risk:** ask about suicidal thoughts when women present with severe premenstrual mood symptoms.
- **Integrate care:** include PMDD screening in sexual and reproductive, school and university health services.
- **Address coping needs:** ask how women manage symptoms and what support they need.

Future Work

- **Confirm PMDD prospectively** in studies from low- and middle-income countries.
- **Study lived experience:** stigma, coping and help-seeking in African settings.
- **Validate screening tools** in African languages.

References



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- Scan the QR code for the full reference list.